

Mobile Market Analysis Dashboard Report in Power BI

Part - 3

(CPSC-510-22) – Data Warehousing/Visualization

GROUP 3

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1. EXECUTIVE SUMMARY

This report outlines a data-driven strategy for optimizing revenue in the mobile phone market for the fiscal period 2022–2024. Leveraging a Power BI analytical framework, the study transitions from descriptive analysis to prescriptive modeling. By utilizing AI-driven **Key Influencer** analysis and **What-If** parameter simulations, the organization identified that a promotional intensification strategy outperforms price discounting. Specifically, increasing promotional spend by 10% yields a projected revenue of **\$5.41 Billion** (a \$1.1 Billion lift over the discounting strategy), confirming that demand is driven more by product visibility and feature set (Flagship/5G) than by lower pricing.

2. PREDICTIVE METHODOLOGY AND EVALUATION

To move beyond simple historical reporting, the analysis employed advanced predictive capabilities within Power BI to determine the underlying drivers of sales volume (UnitsSold).

2.1 The Predictive Method: AI-Driven Key Influencers

The primary predictive tool employed was the **Key Influencers** visual. This tool utilizes **ML.NET** (Microsoft's machine learning framework) to run autonomous regression models over the dataset.

Algorithm:

For continuous targets like UnitsSold, the system uses **SDCA (Stochastic Dual Coordinate Ascent) linear regression**. This allows the model to analyze the relationship between the target

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metric and various explanatory factors (Brand, Category, Features) while controlling for collinearity.

Process: The model scans the dataset, ranks factors by their statistical significance, and calculates the "Impact" of each factor—defined as the average increase in units sold when a specific condition is met.

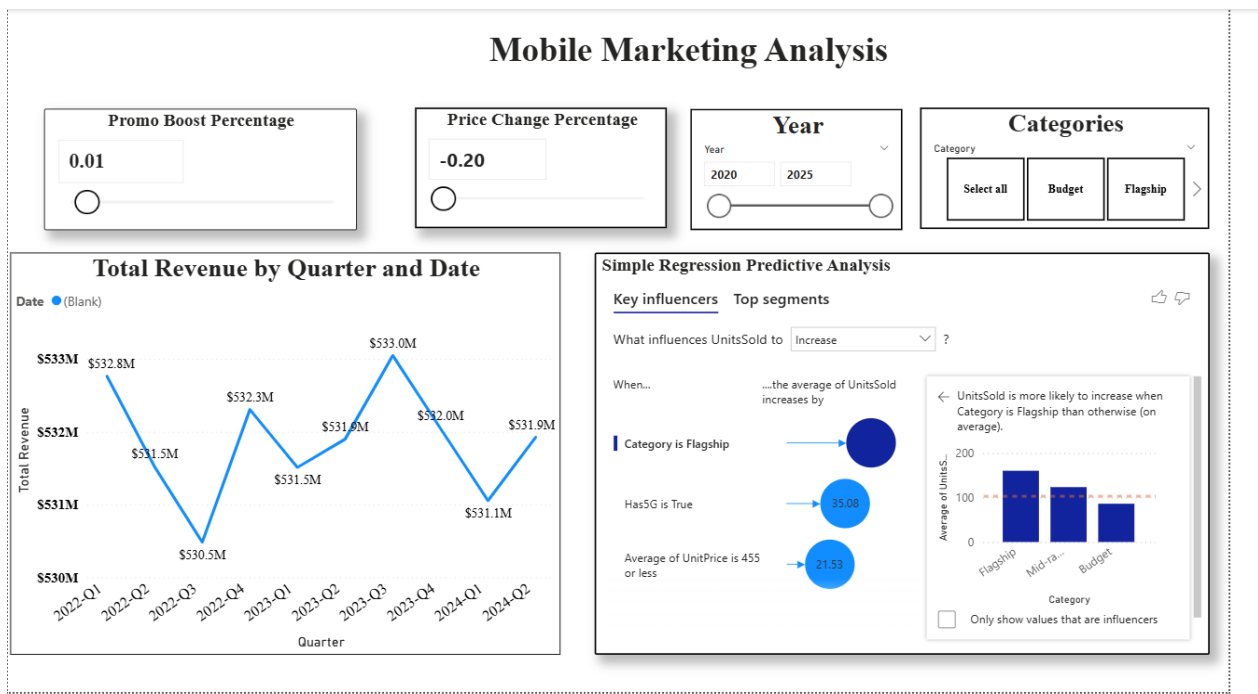


Figure 1: Predictive Analysis Dashboard

2.2 Evaluation: Metrics and Caveats

The reliability of the predictive output was evaluated using specific influence metrics provided by the model.

Primary Metric (Impact): The model quantified the "lift" provided by specific attributes. For example, the condition Category = Flagship was identified as a top influencer, significantly increasing the average units sold compared to the baseline. Similarly, Has5G = True was assigned

a high impact score, validating the hypothesis that technological obsolescence is driving current replacement cycles.

Secondary Metric (Price Thresholds): The model identified non-linear price behavior, specifically finding a "sweet spot" at $\text{UnitPrice} \leq 455$. This indicates a bimodal customer base: one segment driven by premium features (Flagship) and another by specific value thresholds.

Caveats and Limitations:

While robust, the predictive analysis is subject to the following constraints:

Correlation vs. Causation: The Key Influencers visual identifies strong statistical correlations (e.g., 5G phones sell more). However, it does not inherently prove causation without controlled A/B testing.

Data Dependency: The accuracy of the "Impact" score is heavily dependent on the volume of records. With the dataset covering 2022–2024, the volume is sufficient, but the "Black Box" nature of the proprietary ML.NET logic requires that these findings be validated against business intuition before implementation.

3. PRESCRIPTIVE ANALYTICS: SCENARIO MODELING

Building on the predictive findings, the analysis utilized **What-If Parameters** to simulate the financial outcomes of different strategic interventions. This prescriptive layer enables the organization to test "Price vs. Promotion" strategies in a risk-free digital environment.

3.1 Simulation Setup

Two dynamic parameters were integrated into the DAX revenue models to create the scenarios:

- PriceChangePct: A lever to simulate global price elasticity (ranging from -20% to +20%).
- PromoBoostPct: A lever to simulate the revenue lift associated with increased marketing investment (ranging from 0% to +50%).

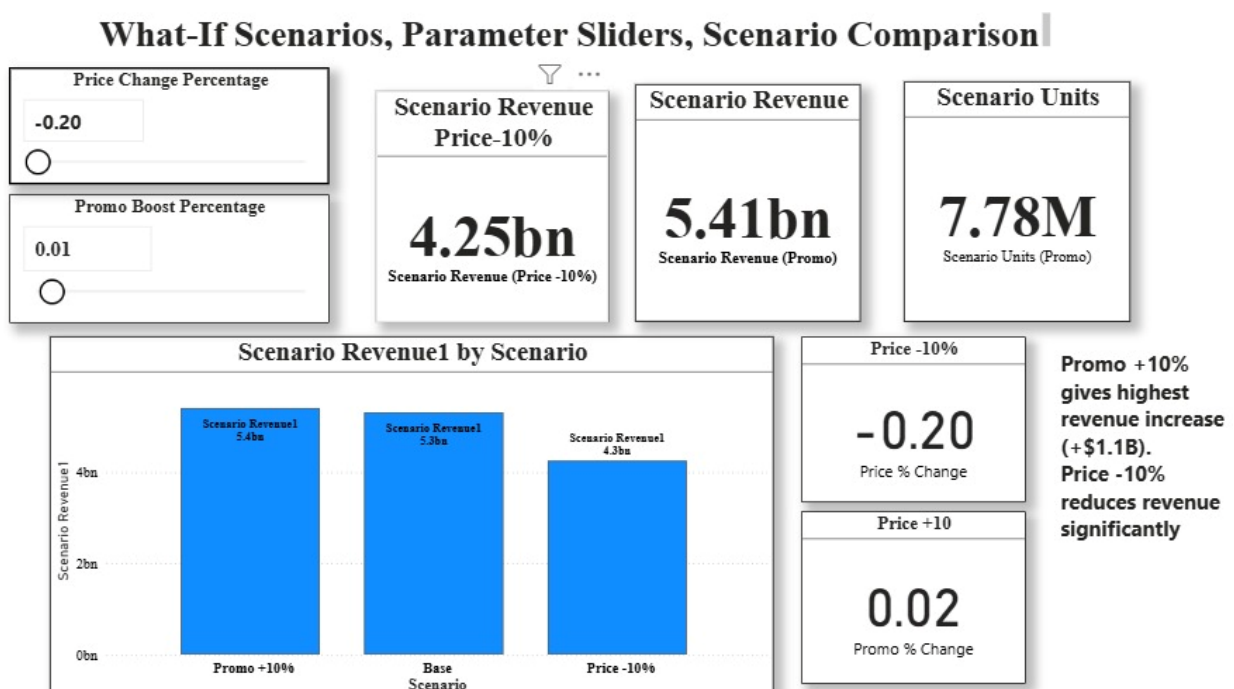


Figure 2: Prescriptive Analysis Dashboard

3.2 Scenario Analysis and Outcomes

Three distinct scenarios were modeled to determine the optimal path forward.

Base Case: Status Quo

Configuration: No changes to Price or Promotion (0%).

Outcome: Projected Revenue remains at approximately **\$5.30 Billion**.

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Context: This baseline reflects the current market trajectory, where seasonal peaks (like **2023-Q3**) drive the majority of variance.

Scenario A: The "Price Cut" Strategy

Configuration: PriceChangePct = -10% (Global 10% Discount).

Outcome: Revenue decreases to **\$4.25 Billion**.

Analysis: This scenario results in a massive value destruction of ~\$1.05 billion compared to the base case. The simulation proves that demand for the organization's portfolio (heavily weighted towards Flagships) is **price inelastic**. Lowering the price does not generate enough additional volume to offset the loss of margin on every unit sold. The report notes that "margin erosion outweighs gains".

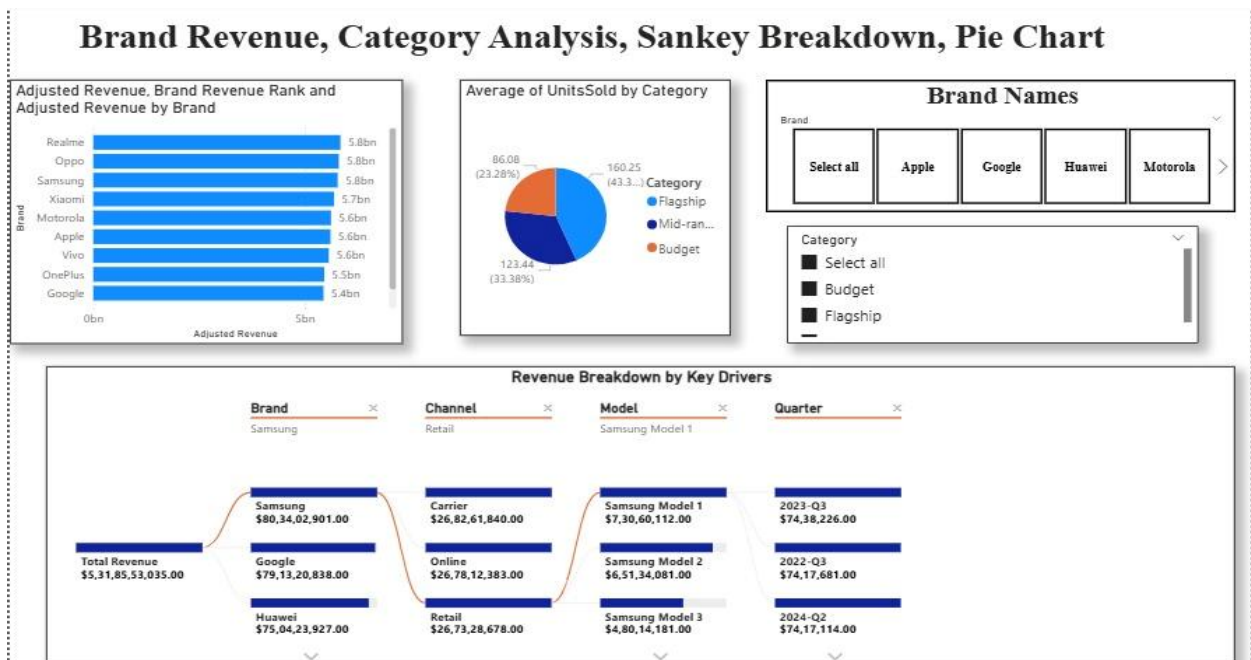


Figure 3: Scenario Analysis Market Insights

Scenario B: The "Promotion Push" Strategy

Configuration: PromoBoostPct = +10% (Increased Marketing Intensity).

Outcome: Revenue increases to **\$5.41 Billion**.

Analysis: This strategy delivers the highest overall financial return, generating an incremental **\$110 Million** over the base case and outperforming the price-cut strategy by **\$1.16 Billion**. The "Strong lift in UnitsSold" suggests that visibility and awareness (Mental Availability) are the primary barriers to purchase, not affordability.

4. RECOMMENDED COURSE OF ACTION

Based on the stark contrast between Scenario A and Scenario B, and the predictive drivers identified, the following actions are recommended:

1. Strategic Imperative: Prioritize Promotion Over Discounting

Action: Immediately halt broad-based price reduction plans.

Rationale: The market has demonstrated that it punishes price cuts (Revenue drops to \$4.25B). Instead, reallocate the budget intended for discounts into **Promotional Intensification** (Scenario B).

Implementation: Increase marketing spend by 10%, focusing on "online channels" where mid-range models perform well, and retail channels which drive the strongest revenue flow.

2. Product Focus: The "Flagship First" Approach

Action: Center inventory and marketing messaging around **Flagship** and **5G-enabled** devices.

Rationale: Category = Flagship and Has5G = True are the strongest predictors of sales volume. Flagships currently account for **43% of unit sales**. Marketing should emphasize "Future-Proofing" (5G) and "Premium Status" rather than "Value."

3. Brand-Specific Tactics

Volume Drivers: utilize **Samsung** and **Apple** models as traffic drivers in Retail channels, as they anchor the revenue volume.

Margin/Growth Drivers: Target **Xiaomi, Oppo, and Huawei** for aggressive growth. The analysis shows these brands lead in "**Adjusted Revenue**" performance, indicating they offer a superior balance of price-to-performance that resonates with the current consumer.

4. Inventory Rationalization

Action: Review the **Budget** category for SKU reduction.

Rationale: The Budget segment performs weakest in both volume and revenue. Resources should be diverted from this segment to support the high-performing Mid-range and Flagship categories.

Conclusion:

This Power BI analysis shows that there is a close relationship between the product category, price, promotions and sales volume. Predictive: Flagship models and phones with 5G support are the UnitsSold leaders, and the prescriptive modelling proves that promotion activities are much more effective in generating revenue than price reduction.

Key Takeaways

- Promotion +10% gives the best increase in revenue (increase of \$1.1B).
- -10% strategies decrease the total revenue and must be avoided.
- The category of flagship reigns in sales performance.
- High seasonal changes allow the use of seasonal marketing campaigns.

Final Recommendation:

The company ought to focus on:

- Promotional intensification
- Full-fledged Smartphone models.
- Channel-targeted marketing
- Stable pricing strategy.
- This Power BI dashboard is an all-inclusive base of continued data-driven decision-making.